Surveillance Over Time:
Automated Licence-Plate Readers on American Commutes, and the Rising Price of Avoiding Them

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October 26, 2026

Abstract

Automated licence-plate readers (ALPR) have spread from tollbooths to residential streets. We measure what that means for an ordinary trip, and how fast it is changing. A continental routing engine models each camera's field of view and can find a route that crosses none. We route 56,133 home–work commutes from all fifty states and the District of Columbia, with and without avoidance, past 186,247 installations: 143,721 mapped by volunteers in OpenStreetMap, less 168 since taken down, and 42,694 more in service that only the registry of the dominant vendor, Flock Safety, records. 78.6 % of commuters pass at least one reader on the way to work. Refusal is still cheap: 80.6 % can reach work past none, and avoiding adds a median of 2.29 min—about two-fifths more than the volunteer map alone implies. OpenStreetMap keeps every edit, so we also rebuild its camera map as it stood on the first of every month since January 2024, node for node, and re-ask the question of each on the same commutes and roads. The map grew 129-fold in thirty-four months, and exposure almost in proportion: the cameras an average commute passes rose 96-fold, and the share of commuters passing one from 2.2 % to 75.6 %. Refusal grew dearer per camera as well as in total: on the first map of 2024 95 % of exposed commuters could reach zero, for a median of 0.8 min; on today's map 79 % can, for 3.2 min, and 75 % once the vendor's registry is added, for 3.9 min. What a commute passes, and what declining costs, are functions of a map that is still filling in; we say when ours was drawn.

Keywords: automated licence-plate recognition; surveillance; route choice; volunteered geographic information; OpenStreetMap; edit history.

1 Introduction

Automated licence-plate readers (ALPR) photograph every passing vehicle, resolve the plate, and retain the read [24]. Unlike a traffic camera they are not triggered by an offence, and unlike a patrol officer they do not tire. Fixed, pole-mounted units have spread from limited-access highways into residential neighbourhoods, arterial roads, and the entrances of private subdivisions [49, 68].

The public debate about this expansion is conducted in the vocabulary of policy: retention windows, warrant requirements, audit logs, data-sharing agreements. What that vocabulary lacks is a measurement of the thing being regulated. How much of this does a person actually drive through on the way to work on an ordinary Tuesday? And if they would rather not, is declining a practical option or a theoretical one?

This paper answers both, and does so by building the instrument the question implies. We construct a routing engine that treats ALPR fields of view as first-class geometry and can be asked

for a route that avoids them. Running each sampled commute through it twice—once normally, once avoiding—yields the exposure and the price of refusing it, for the same trip, on the same network, at the same speeds.

The second quantity is where the paper earns its keep. A privacy protection that costs four minutes is a feature. One that costs forty is available only to people with forty minutes to spare. We therefore report the price of refusal as a distribution rather than a point: for most commuters it is a few minutes, and for a tail it is not.

The third question follows from the data source, and it is the one this paper is about. The cameras are mapped by volunteers, and the map has grown 129-fold since January 2024. OpenStreetMap keeps every edit, so the measurement can be repeated on the map as it stood at any past instant. We repeat it on the first of every month for thirty-four months: the same commutes on the same roads, re-scored against each month's cameras and re-routed against each quarter's. How much surveillance a commuter drives through, and what declining it costs, turn out to be functions of how much of it had been mapped—and both have risen since, exposure on all but one of the monthly maps and the cost on every routed map from July 2024.

A volunteer map is also a lower bound, and for the present it can be checked against the vendor's own records. Flock Safety's device registry as it stood in December 2025 has been published; merged with the map, it adds 42,694 in-service readers in the fifty states and the District that no mapper had reached, nearly a third again. The current figures in this paper rest on that merged map. The trend, which a registry without dates cannot join, rests on the map alone, and the difference between the two is itself a measure of what the volunteer map still misses.

1.1 Contributions

- **A directional exposure model at continental scale.** Cameras are wedges, not circles. 98.6 % of mapped North American ALPR installations carry a bearing tag, making this feasible nationally; proximity counting would charge drivers for cameras aimed at the opposite carriageway.
- **Avoidance routing with no forked router.** Camera fields of view—the map's 160,230 cones, merged where they overlap into 138,046 polygons, and with the registry's 42,811 discs 160,984—are compiled into a stock GraphHopper spatial index at import time, so an ordinary per-request custom model can reference them by name.
- **A commute-weighted sample.** 56,133 real home–work pairs in all fifty states and the District of Columbia, drawn from LEHD LODES with probability proportional to worker count and weighted to each state's commuters.
- **The camera map at any instant, from the edit history.** Every version of every node ever tagged as a reader, extracted once from OpenStreetMap's full-history dump and validated node for node against the public history server, gives the map on the first of every month since January 2024 and the day each camera entered it. The map's coverage is checked against the counts 1,096 police departments publish, of which it holds a median 100 %.
- **The map, merged with the vendor's own registry.** Flock Safety's device records as they stood in December 2025, paired one-to-one with the map within 50 m: 89,363 registry devices stand on a mapped node, and the 42,694 in-service plate readers in the fifty states and DC that the map lacks enter the current picture as 60 m discs, the registry recording no bearing. They raise the price of refusal by two-fifths. The current figures rest on this merged map; the trend, which an undated registry cannot join, on the map alone.
- **A thirty-four-month trend on a fixed sample.** The same commutes on the same roads

against thirty-four monthly maps, re-scored for exposure on each and re-routed for avoidance on each quarter's: exposure rose almost in proportion to the map, and refusal grew dearer per camera as well as in total.

2 Related work

This paper sits where three bodies of work meet: research on ALPR as a policing technology, on volunteered geographic data and its biases, and a small practice of routing around surveillance. We take them in turn, and then state what is, to our knowledge, new here.

ALPR as a policing technology. Criminological research on ALPR has mostly asked whether it works: Lum et al. [48] tested ALPR-equipped patrols in a place-based randomized design, Lum et al. [50] review the impact studies since, and Shjarback and Sarkos [66] evaluate Atlantic City's fixed readers on every entrance and exit to the city. Sociological and legal work asks what the technology does to policing itself—plate readers lowering the threshold for inclusion in police data [5], the automated suspicion they enable [22, 41], and Flock's network as an infrastructure through which policing crosses jurisdictions [56]—and the legal status of the location records they produce is unsettled [6, 24]. None of this work measures how much ALPR surveillance an ordinary trip passes, or how fast that is changing.

Volunteered geographic data. The camera map is volunteered geographic information [33], whose quality has been assessed against authoritative data since OpenStreetMap's early years [36] and whose completeness is known to vary with place and development [38]. For surveillance specifically, the Atlas of Surveillance records which agencies use which technologies but not where devices stand [18]; DeFlock supplies the locations [10]. Coverage bias is the central threat to every result built on such data, and we know of no earlier measurement of it for ALPR; we compare the map with the camera counts police departments publish in Flock's transparency portals (Section 8.2), and its own history shows how far it was from complete on any earlier date.

Exposure along the way. That exposure to a place-bound feature depends on where people go and not only on where they live is the uncertain geographic context problem [44]; measuring surveillance along the commute rather than at the home address is an answer to it, and the unit of analysis throughout is the trip.

Routing around surveillance. A route that avoids cameras is an older idea than ALPR: in 2001 iSee computed "paths of least surveillance" around Manhattan's closed-circuit cameras [40, 65], of which a civil-liberties survey had counted some 2,400 visible from the street [57]. Several services now route drivers around mapped ALPR, some using each camera's field of view [3, 11, 13, 25], and in network measurement Edmundson et al. [15] characterize, and try to avoid, Internet paths through surveillance states. None publishes a measurement of how much surveillance ordinary trips pass or what avoiding it costs; the routing engine here is the instrument for that measurement, not its contribution.

What is new here. To our knowledge this paper is the first to measure, nationally and on real commute flows, how many ALPR fields of view commuters pass and what avoiding them costs; the first to reconstruct the crowdsourced ALPR map from OpenStreetMap's edit record at monthly

resolution and follow a fixed sample of trips across it; and the first to check that map's coverage against the camera counts police departments publish.

3 The instrument

The measurement needs a router that treats "avoid these 160,984 polygons" as a cost rather than a post-hoc filter. GraphHopper 11.0 [35] supplies it without a fork: it resolves a directory of custom areas into a spatial index *at graph import time*, and—the documentation is ambiguous, so we verified it—an area registered that way can be referenced by name from a *per-request* custom model:

```
{ "priority": [ { "if": "in_alpr", "multiply_by": 0.01 } ] }
```

Shipping 160,984 polygons in every request is not viable; naming them is. The cost is that a new cone set needs a full re-import, about two hours for North America. Two settings follow. Contraction Hierarchies [29] are disabled, since they bake edge weights into the prepared graph and a per-request avoidance strength needs them mutable; landmarks, the ALT algorithm of Goldberg and Harrelson [31], serve instead. And the avoiding profile shares the plain profile's landmark preparation, which is valid only because avoidance multiplies priority strictly downward, so the landmark distances remain the lower bounds ALT requires. The map's cone geometry also serves TRAMES, a fork of the OsmAnd navigation client [62] that routes around the cameras on the phone itself: its offline maps tag each stretch of road between junctions that a cone touches, as the study's router weighs them, and OsmAnd's own router penalises those stretches with the same multipliers, by default 0.01. A public instance of this engine served the app's optional online mode until 29 August 2026; the study's routing server binds to localhost and is not internet-facing. The app avoids the volunteer map alone: the registry's readers that this paper's current figures count (Section 4.2) are not in it, and Section 7 measures what that leaves on a route planned around the map alone.

4 Data

All sources are public and key-free.

4.1 ALPR installations

Source: OpenStreetMap [59], queried through the Overpass API [58] for nodes tagged `man_made=surveillance` with `surveillance:type=ALPR`, matched case-insensitively (19 readers are tagged `alpr`)—the corpus surfaced by the DeFlock project. Retrieved through DeFlock's own Overpass server, the public one being at capacity that day, in tiles of at most 4° over five regions (continental US, Alaska, Hawaii, Canada, Mexico), and deduplicated by OSM node id. The map in this paper is the snapshot of **2026-10-21: 144,692 unique camera nodes**, 143,721 of them in the 50 states and the District of Columbia; of the rest, 942 stand in Canada, 24 in Mexico and five in Honduras, which the Mexican region's box reaches. With the registry's readers (Section 4.2) the merged map holds 186,247 installations in the states and the District (Figure 1). Two earlier snapshots, taken 2026-07-24 and 2026-09-22, held 120,838 (120,271 in the states) and 142,991 (142,255); the second includes 1,181 nodes added in error and since deleted. Section 4.4 reconstructs the map on the first of every month since January 2024, and says how those nodes are handled.

A fetch that fails over between Overpass mirrors has to check the mirror's database date: on the first attempt one regional query was answered from a mirror four months stale, and roughly

Table 1: Directional and vendor tagging in the first 2026 snapshot and the study's. The direction-key rows overlap (392 and 422 nodes carry both keys). Vendors are matched as case-insensitive substrings of the first of the four vendor keys present, in both columns.

	2026-07-24		2026-10-21	
Property	Count	Share	Count	Share
Camera nodes	120,838		144,692	
Any direction tag	119,249	98.7 %	142,730	98.6 %
via `direction`	117,356	97.1 %	140,805	97.3 %
via `camera:direction`	2,285	1.9 %	2,347	1.6 %
Multi-head (`;`-separated)	7,014	5.8 %	9,552	6.6 %
Arc-range form (`199-269`)	4,806 tokens	—	4,557 tokens	—
`operator` tagged	22,354	18.5 %	24,302	16.8 %
Flock Safety	99,650	82.5 %	115,821	80.0 %
Motorola Solutions	5,993	5.0 %	7,698	5.3 %
Genetec	2,876	2.4 %	3,964	2.7 %
Leonardo	1,020	0.8 %	1,332	0.9 %
Axis Communications	1,052	0.9 %	2,317	1.6 %
No vendor tag	5,811	4.8 %	7,022	4.9 %

13,600 cameras silently reverted to old tags with nothing reported as failed. The fetch now rejects any answer older than 48 hours, and every tile of the snapshot used here falls within a 19-minute window.

Three tagging realities shaped the parser. Mappers write the bearing under plain `direction`, not the `camera:direction` the wiki documents [61], by a factor of 60; reading only the documented key would discard 98 % of the bearings. Explicit arcs are real and mostly 45°: 4,557 range values on 4,302 nodes, 3,746 of them (82 %) spanning exactly 45°, which is what justifies 45° as the default span for the 97 % of cameras that give a bare bearing—the default is inherited from the data, not assumed. And multi-head units exist (`320;190`, `0;72;144;216;288`): 9,552 nodes, typically watching both carriageways, so the parser emits one wedge per head rather than silently taking the first, which would make a camera that sees everyone look like one that sees half.

4.2 Flock Safety's device registry

For the present, the volunteer map can be supplemented with the vendor's own records. An independent researcher has published a map of Flock Safety's device registry as it stood in December 2025 [51, 55], which we retrieved on 25 September 2026 (the published file was unchanged on 21 October). Its 335,701 records, 334,525 of them in the fifty states and the District, give each device's position, type (a *Falcon* reads plates; other types record video or sound), status (in service, planned or decommissioned), a label that often names the road and the traffic watched (`#02 Pacific Pointe (Outbound)`), the date its record was created, and a `rotationAngle`, which is not a bearing: on the 73,649 paired devices whose mapped node carries one surveyed bearing, the two disagree by a median of 90°, as two independent angles would, and agree within 20° for 7.9 %. No direction is derived from it.

Each registry device is paired with at most one mapped node, and each node with as many devices as it has heads, within 50 m, nearest first, the two sides' vendor and capability required to agree: 89,363 pairs, covering 64.6 % of the registry's 125,400 in-service plate readers (126,389 before

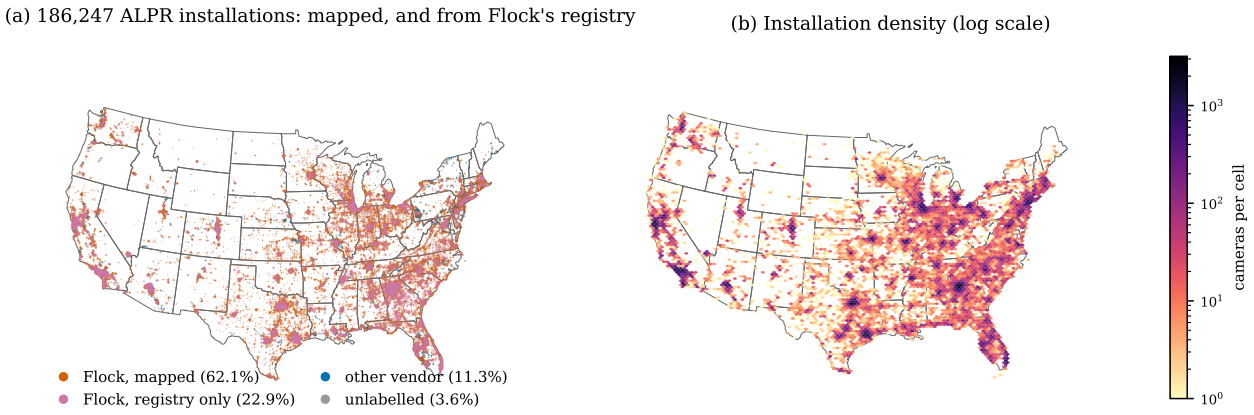


Figure 1: ALPR installations in the contiguous United States: those mapped in OpenStreetMap on 21 October 2026, and the in-service readers in Flock Safety's December 2025 registry that the map lacks, less those confirmed removed since. (a) By source and vendor, the registry's readers drawn on top: they show where the volunteer map falls short. (b) Density of both, on a log scale. The visible structure is partly deployment and partly *recording*: see Sections 6.3 and 8.2. Camera data © OpenStreetMap contributors and Michael [55].

the 989 found taken down, below), at a median separation of 5.6 m. 55,244 mapped nodes have no registry device within reach—another vendor's camera, or one the registry does not carry—and 44,397 in-service plate readers in the registry have no mapped node. We drop 1,586 of those whose records place them at a coordinate shared by many devices (a building's address rather than a pole, 1,473), name them as Flock's own or test units, say they are retired, or duplicate another record, and add the remaining **42,811** to the map: 42,694 in the fifty states and the District, 117 in Puerto Rico and the Virgin Islands, where no sampled commute runs. Planned, decommissioned and removed readers, and devices that do not read plates, stay out. The registry adds only Flock's devices, so on the merged map Flock's share of installations rises from 81 % to 85 %; other makers' unmapped cameras remain unmapped.

A status from December 2025 is ten months old at the cutoff. Where a reader the registry lists as in service is known to have been taken down by 21 October 2026, we mark it removed: 989 records, 417 of them in 32 places where the operating agency or the local press reported the removal, each matched to the agency's own readers by their labels and count, and 572 whose mapped node, paired with the record in September, was deleted by a mapper who gave the camera's removal as the reason. Appendix A lists the places and their sources, and the repository every record with its source. The merged map also leaves out the 168 mapped nodes that stand on a removed reader; the map alone keeps what OpenStreetMap shows.

These are not recent cameras the map has yet to reach: 26,051 of the 42,811 carry the registry's bulk-import date of 26 March 2024, so were on record by then, and only 2,100 were recorded in the second half of 2025. A record's creation date is not an installation date—a device is recorded while planned—and the registry gives no date of removal, so it cannot be reconstructed as it stood on an earlier day. The trend of Section 6.4 therefore rests on the map alone.

4.3 Cone construction

Each camera head becomes a circular sector: apex at the node, bisector along the tagged bearing, span 45° (or the tagged arc), radius 60 m. The radius must reach across the carriageway from a set-back pole; it was chosen on that reasoning and then swept over $\{30, 45, 60, 90\}$ m (Section 8.4). Measured exposure is only mildly sensitive to it, but the avoidance result is not robust above the assumed value.

All wedges are unioned into a single MultiPolygon of **138,046 parts**, published as one feature with id `alpr`. Overlapping wedges merging is correct and intended: the question is *is this road segment observed*, not *by how many devices*.

A camera with no bearing receives no cone and contributes no exposure: 1,962 nodes carry no direction tag and 70 an unparseable one, 2,032 of 144,692 (1.4 %). A full circle would overstate a camera that watches one approach; excluding them understates by at most that share, and makes measured exposure a lower bound on the map's own terms as well as on the world's. Thirty-four nodes give a full-circle arc (`0-360`), and each becomes a disc of the cone's radius. The vendor is read from four inconsistently spelled keys by case-insensitive substring (115,821 Flock cameras, against 111,355 by exact match); it matters only for the vendor split of Section 6.1, and the avoidance model targets the all-vendor union throughout.

The registry's readers carry no bearing at all, and excluding 42,811 cameras would discard the registry. Each receives instead a **disc of the cone's radius**, 60 m in every direction, the rule the published device map applies to any device of unknown heading. A disc watches both carriageways where a Falcon watches one, so it can overstate a reader's view; Section 5.2 scores the discs a second way, from the direction named in the device's label, and Section 6.1 reports both. The merged set is the map's wedges, less those of the 168 nodes that stand on removed readers (137,898 parts), plus the discs: **160,984 parts**, one feature with id `alpr_merged`, imported as a graph of its own.

4.4 Camera maps of earlier dates

OpenStreetMap keeps every edit, so the camera map can be reconstructed as it stood at any past instant. The public Overpass server answers a query carrying a `[date:...]` qualifier from its history: nodes deleted since are restored, nodes added since are absent, and every tag has its value from that day. Asked for Georgia at the July snapshot's timestamp it returned 9,742 nodes, the snapshot's own count to the node. History queries for the whole continent time out on the public server, so a date is fetched state by state and merged by node identifier; DeFlock's own Overpass mirror keeps no history and cannot serve them. They are heavy, and after one complete date the public server began refusing our connections.

Every other date therefore comes from OpenStreetMap's full-history dump [60] (164 GB, 12 October 2026, checked against its published MD5), read once. Two passes collect every version of every node that was ever tagged as an ALPR—177,541 nodes and 264,990 versions, including the later versions that deleted or retagged them—and the map at any instant is then a filter over that file, with Overpass's semantics: each node's latest version at or before the instant, kept if it was visible and still tagged. On the one date both sources cover, 1 July 2025, they agree exactly—22,690 nodes, the same identifiers, the same tags, the same coordinates—and on 1 January 2026 the filter reproduces an earlier, independent pass over the dump node for node (63,526). The same file gives the day, to the second, on which each camera entered the map, and a monthly count of the map that needs no aggregating service.

We reconstruct the map on the first of every month from January 2024 to October 2026, thirty-four maps, and set them beside the study's snapshots of 24 July, 22 September and 21 October 2026.

Cones are built from each exactly as from the current map (Section 4.3). The share of mapped cameras carrying a bearing, and so receiving a cone, was about 90 % in 2024 and 99 % in 2026: the earliest maps are slightly less complete even on their own terms. Exposure is measured on every monthly map; avoidance is routed on the first of every quarter (Section 5.3).

One correction applies to these maps. On 18–20 September 2026 one contributor added 1,181 nodes as plate readers that were mostly airport surveillance cameras—1,040 in Chicago, around O'Hare and in the Loop, and 141 in Bismarck, North Dakota, none of them Flock's—and DeFlock's reviewers had deleted or retagged every one by 19 October. They are in the 22 September snapshot as retrieved and, 642 of them, in the map of 1 October; both enter the trend without them, and the list is in the repository.

One caveat governs everything built on these maps. OpenStreetMap records when a camera was *mapped*, not when it was installed: an installation date is present on 49 of 144,692 nodes. A camera first mapped in 2026 may have been on its pole since 2021. A past map is therefore what a study run on that day would have seen, and the trend of Section 6.4 describes how the measured picture changed as volunteers filled in the map—most of it after DeFlock was founded in October 2024 [10]—not how fast cameras were deployed.

4.5 Road network

Geofabrik North America extract [30], 19.5 GB, with OpenStreetMap's roads as of 20 October 2026, held fixed for every camera map, so that the trend of Section 6.4 isolates the camera data: every unavoided route is the same route on every date. Imported with `ignored_highways = footway, construction,cycleway,path,steps`. `track` is deliberately retained: unpaved tracks are plausible avoidance routes, and excluding them would bias the cost estimate downward by removing options a determined driver has.

4.6 Commute sample

Source: LEHD **LODES8** [73], file `od_main_JT00`: all jobs with both ends in-state, built from unemployment-insurance wage records linked to employer establishments. These are administrative counts of real jobs, not survey estimates. The year is 2022 for 49 of the 51 jurisdictions; LODES has not published 2022 for Michigan, which uses 2021, or for Alaska, whose latest year is 2016. LODES8 codes every year against 2020 census geography, so an older year still names current tracts.

Sampling design carries more weight here than sample size. Uniform random points would over-represent empty rural space where nobody commutes and no cameras exist, understating exposure badly. Instead each origin–destination pair is drawn **with probability proportional to the number of workers actually making that trip**, so the distribution approximates what an average *commuter* experiences rather than what an average *square kilometre* contains.

Block-level LODES geocodes are aggregated to **tract** (first 11 digits), finer than the routing's effective precision, and each tract is routed from its 2024 Census Gazetteer internal point [72] (`INTPTLAT/INTPTLONG`), which is guaranteed to fall inside the tract polygon. Same-tract commutes are dropped: origin and destination centroids coincide, so there is no route to measure. Georgia alone yields 1,199,250 distinct tract pairs covering 4,190,313 workers; across the 50 states and the District of Columbia the frame holds 132,822,679 commuters over 34,420,305 tract pairs.

Connecticut needed one repair: it replaced its counties with planning regions in 2022 and every tract identifier changed, so LODES and the Gazetteer name the same tracts differently and the state would silently drop out; a crosswalk on the six-digit tract code matches all 879 tracts in the flows.

1,200 pairs are drawn per jurisdiction, 61,200 in all, with replacement from a single seeded stream (seed `20260725`), giving **58,421 distinct commutes**. The fifteen states of the study's earlier rounds are drawn first, so their 17,904 commutes are reproduced exactly. The sampled pairs represent 2,842,747 workers; the median pair carries 16 and the largest 2,832.

A pair drawn more than once is routed once, since its route is the same, but it stands for every draw: **each commute carries its draw count as a weight**. In the fifteen large states this is immaterial—0.53 % of draws are repeats—but in small ones it is not. Wyoming's 1,200 draws hold 832 distinct pairs (30.7 % repeats), Alaska's repeat a quarter of the time and the Dakotas' nearly a fifth, and one Vermont pair was drawn nine times. Collapsing repeats at equal weight, as this study's earlier rounds did, under-represents precisely the heaviest flows; keeping the counts restores probability-proportional-to-workers sampling inside every state.

The pairs are what LODES records, not trips anyone has been seen to make. The median routed commute is 23.6 km, but one pair in eight is more than 100 km apart by road (12.8 % of commuters, weighted) and one commuter in a hundred more than 470 km: LODES places each job at an employer establishment, imputed where an employer has several, and pairs that far apart are unlikely to be driven daily. They are kept, since any cut-off would be arbitrary, and their weight is reported where it matters (Sections 6.1 and 8.6).

5 Method

5.1 Two routes per commute

Only the trip to work is routed. A camera watches one direction of travel, so the journey home passes a different set of cameras; it is not measured here, and no figure in this paper should be doubled to stand for a day. Each sampled pair is routed twice against the same graph: a **baseline** with no custom model, and an **avoided** route applying `priority: [{if: "in_alpr", multiply_by: 0.01}]`. Both disable CH, which per-request custom models require.

Both use GraphHopper's stock car model, which weighs distance as well as time (`distance_influence` 90: a route 1 km shorter is preferred if it costs under 90 s more). The baseline is therefore the router's preferred route rather than strictly the fastest, and for 4.0 % of commuters on the merged map (4.6 % on the map alone) the avoiding route—in every such case the longer one—is the quicker, by a median of 0.8 min. Extra time is reported as measured, negative values included; clipping them at zero leaves every median unchanged and raises the mean extra time from 7.33 min to 7.40 min (on the map alone, from 5.63 min to 5.71 min).

0.01 is the application's *strongest* avoidance setting and, since version 1.2.7, its default: a watched road costs a hundred times its usual weight, close to but short of prohibition. Results therefore describe the **strong end of both the benefit and the cost** of avoidance as offered, which is what the application does unless its user chooses a lighter setting. This is deliberate: it establishes whether meaningful avoidance is possible at all, and what it costs near the limit.

5.2 Exposure is directional, not proximate

A camera counts as having seen the driver *only if the route line intersects its cone.* Passing 20 m from a camera aimed the other way is not exposure. Implementation is an STRtree over the cone parts for candidate lookup, then an exact intersection test.

Two simplifications remain, and they pull in opposite directions: route geometry is a polyline while a vehicle has width, which understates exposure, and a cone occluded by intervening buildings is still counted as full sight-line, which overstates it.

A registry disc has no direction, so a route that crosses one counts it. The device labels supply a second, directional, reading: 25,155 of the 42,811 name a cardinal direction of travel (*northbound*, ...) and 4,047 say inbound or outbound. Under the *directed* scoring a cardinal disc counts only if the route crosses it travelling within 60° of the named heading, taken from the route's bearing where it passes nearest the device; the other discs count as before. The two scorings bracket the registry's contribution: every reader on both carriageways, or every reader on the one its label names.

5.3 Earlier camera maps need no new baselines

The unavoided route does not depend on the cameras. Routed two months apart against graphs carrying different camera maps, all 17,580 commutes of the study's earlier rounds took identical baseline routes, to the metre and the second. Exposure on an earlier date is therefore a re-scoring of each saved route against that date's cones, with no routing. The merged map is treated the same way: the saved baselines re-scored against its cone set, and avoidance routed against a graph carrying that set for every commute the merged map exposes.

Avoidance does need routing against that date's cameras, and two properties keep it tractable. First, one graph serves every date: each date's cone set is imported as its own custom area (`alpr_2025_07_01`, ...), and each request names the one it avoids —an import of about two hours serves five or six dates rather than one, checked on a test graph to route each area independently. Second, only commutes that pass a camera on that date are routed. A route that crosses no cone is already the avoiding route, since the custom model only makes cone edges dearer and nothing can undercut it; this was checked rather than assumed, and all 3,741 such commutes of the fifteen-state run received an identical avoiding route. The same shortcut serves the main run.

Re-scoring uses route geometry stored at five decimal places, about a metre. Against live scoring on the current map it moves 1.6 % of commutes by one camera, in the two directions about equally, changes whether 5 of 17,579 (0.03 %) pass any camera at all, and leaves national totals within 0.1 %. On the full sample the re-scored national figures match the live ones to the precision reported (75.6 % passing a camera, a mean of 5.17). Within the trend every date, the current one included, is scored the same way. Exposure is re-scored on every monthly map; avoidance is routed on the first of every quarter, eleven dates from January 2024 to July 2026, and on the 21 October 2026 snapshot itself. The snapshots of 24 July and 22 September 2026 enter re-scored only.

5.4 Statistical reporting

- **Medians lead, means accompany.** Both distributions are heavily right-skewed; a minority of long commutes through dense corridors pull the mean well above the typical experience.
- **National figures are weighted by commuters.** Every jurisdiction was sampled at the same size, so an unweighted pool would count Wyoming's 198,447 in-state commuters as heavily as California's 16,821,903. Each routed commute carries its state's commuter total from the LODES frame, shared among that state's commutes in proportion to draw count; medians are weighted quantiles, and state figures need only the draw counts. Weighting alone moves the share of commuters passing a reader in the fifteen states of the earlier rounds from 82.3 % to 84.3 % on the merged map (from 78.8 % to 81.4 % on the map alone), because the populous states are the more surveilled ones.
- **Intervals from a bootstrap stratified by state** [16]: percentile, 2,000 replicates, each resampling every state at its own size and re-deriving its weights, as the sample was drawn. Commutes from one county share corridors and cameras; resampling whole origin counties

within states instead [23] widens the headline intervals by a factor of 2.3 to 3.4—78.6 % becomes [77.3–79.5], the median 2.29 min [2.12–2.54]. Where the two disagree, both are shown.

- **Differences between the two maps are paired.** Both runs route the same commutes, and the bootstrap draws the same replicates for each, so a difference is bootstrapped replicate by replicate rather than read off two separate intervals.

5.5 Execution

Twelve concurrent workers against the local routing server, past which contention outweighs added concurrency. A commute whose route fails is tried once more and then dropped: **56,133 of 58,421 commutes routed (3.9 % dropped)**. Nine in ten failures are points the router cannot snap to any road—a tract's internal point deep inside a national park, a reservation or a wilderness tract—and the rest are genuinely unconnected, such as commutes between Hawaiian islands. Five states lose more than a tenth of their sample (Alaska 33 %, Hawaii 24 %, Montana and Wyoming 21 %, New Mexico 14 %); the median state loses 2.0 %. Those five hold 1.6 % of the nation's commuters, so the national figures barely register the loss, but their own estimates describe the routable, probably less remote, part of each state. A wider road search could snap more of them, but in a test it moved 5 of 100 already-routed commutes to a different road, and the trend of Section 6.4 pairs every saved route with new ones; consistency was kept over coverage. Commutes are processed in seeded random order, so any prefix of the output is a near-uniform sample of the whole. Against the merged map the 38,034 commutes it exposes were routed again, none failing once the 61 that failed at first were repeated; the two runs' unavoided routes are identical to the metre and the second, and the merged camera count re-scored from stored geometry matches the router's on every one.

6 Results

6.1 Exposure on the unavoided commute

78.6 % of American commuters pass at least one ALPR on the way to work. The median commuter passes three and the 90th percentile sixteen, and aggregate exposure is 0.116 cameras per kilometre driven (Table 2). The 99th percentile, forty-two, belongs mostly to the long pairs of Section 4.6: the commuters there travel a median 300 km one way, and among the 87 % whose pair lies within 100 km the 99th percentile is thirty-one. Against the map alone, the registry's readers add 1.04 encounters to the average commute [1.00–1.07], a fifth more, and 3.0 points [2.8–3.2] to the share passing any reader. They mostly thicken what the map already shows: the commuters the map alone exposes pass a median of five readers instead of four, while 3.1 % [2.9–3.3] pass one only once the registry is added.

A disc sees both carriageways, where a Falcon watches one. Scored instead by the direction each reader's label names (Section 5.2), 77.8 % [77.4–78.3] of commuters pass a reader and the mean is 5.81: 38 % of the encounters the registry adds are with readers whose labels say they face the other way. The share passing any reader, and the avoidance results, move by less than a point; scored by direction, the avoiding routes reach zero for 80.9 % of commuters, against 80.2 % for the same stored routes scored with discs.

By vendor, on the map alone, where every cone comes from a surveyed bearing, Flock Safety holds 81.1 % of the installations that contribute cones in the fifty states and the District but supplies 67.2 % of commuter exposure; per unit deployed, other makers' cameras generate 2.27 times as many encounters, being multi-headed more than six times as often (23.3 % against 3.7 %) and standing

Table 2: Exposure and the cost of avoidance for American commuters, on the merged map (OpenStreetMap's cameras of 21 October 2026 with the in-service readers only Flock Safety's registry records) and on the map alone: 56,133 routed commutes in the 50 states and the District of Columbia, weighted to the 132.8 million in-state commuters they represent. Medians are weighted; intervals come from a bootstrap stratified by state. Median commute 23.6 km / 23.4 min.

	Merged map		Map alone	
	Median [95 % CI]	Mean	Median [95 % CI]	Mean
Unavoided route				
Cameras passed	3 [3–3]	6.21	3 [3–3]	5.17
Commuters passing ≥1	78.6 % [78.2–79.1]		75.6 % [75.2–76.1]	
Commuters passing ≥5	42.0 %		36.9 %	
Commuters passing ≥10	20.8 %		16.6 %	
90th / 99th percentile	16 / 42 cameras		13 / 35 cameras	
Exposure rate (per km)	0.116 cameras		0.097 cameras	
ALPR-avoiding route				
Cameras passed	0 [0–0]	0.36	0 [0–0]	0.27
Reduced to zero	**80.6** % [80.1–81.2]		84.1 % [83.6–84.6]	
Extra time (min)	**2.29** [2.21–2.36]	7.33	1.64 [1.58–1.70]	5.63
Extra distance (km)	0.86 [0.82–0.89]	3.36	0.58 [0.55–0.61]	2.60
Avoidance overhead	9.65 % [9.41–9.92]	17.11 %	6.82 % [6.55–7.07]	13.48 %
Minutes per camera evaded	0.83 [0.82–0.85]	1.37	0.76 [0.75–0.78]	1.30

more often beside principal arterials and freeways (44 % within 30 m of one in the federal traffic inventory [21], against 26 % of Flock's). Where the operator is tagged, police departments run about two-thirds of both. The registry adds only Flock's readers, and as discs, so a per-unit comparison on the merged map would measure the disc rather than the vendor; on that map Flock holds 85 % of all installations, against 81 % of the map's. Each commute's saved route re-scored against each vendor's cones gives the split without new routing; the avoidance model does not distinguish vendors.

6.2 The cost of refusal

80.6 % of American commuters could be routed to zero ALPR exposure: the fifth whose usual route passes no known reader, and three in four of the rest (75.4 % [74.7–76.0], Table 3). Over all commuters the avoiding route adds a median of 2.29 min and 0.86 km, an overhead of 9.65 %; for commuters whose usual route passes a camera it adds a median of 3.91 min [3.81–4.01]. The median marginal price is 0.83 min per camera evaded.

The registry's readers make refusal dearer on every measure. On the same commutes, against the map alone, the share able to reach zero falls by 3.5 points [3.1–3.8], the median extra time rises by 0.65 min [0.60–0.69], about two-fifths, and the price of evading a camera by 0.07 min [0.06–0.08]. For the commuters the map alone already exposes the median detour grows from 3.16 min to 4.10 min, and 4.5 % of all commuters [4.2–4.8] who could reach zero against the map cannot once the registry is added. Extra time rises for a third of commuters, is unchanged to within three seconds for three in five, and falls for one in twenty, where the router, finding its old detour watched too, more often keeps a camera (43 % of those routes do, against a quarter of all exposed commuters' routes).

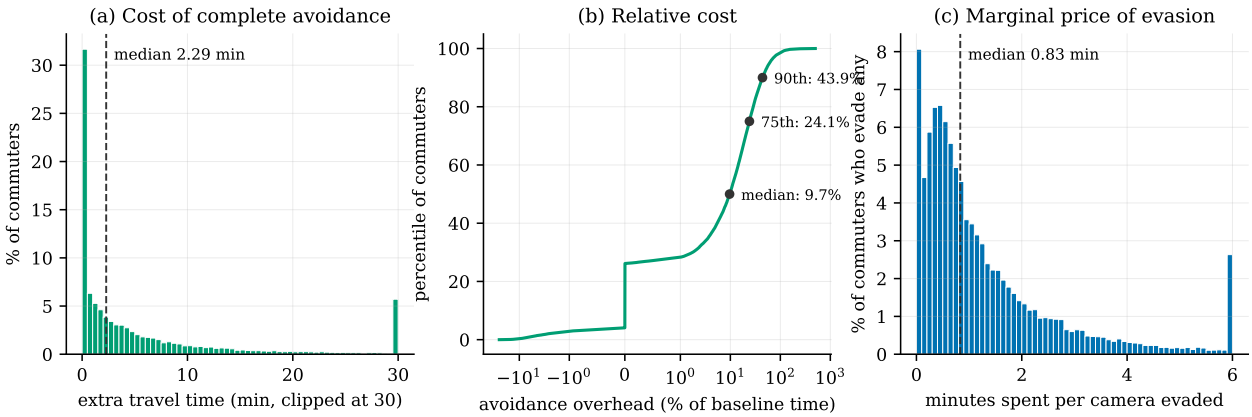


Figure 2: The price of complete avoidance, as shares of commuters. (a) Absolute extra time; (b) overhead as a percentage of baseline, by percentile; (c) marginal minutes per camera evaded. All three are heavily right-skewed—a small tail of commutes pays a great deal, while the typical commute pays almost nothing.

For most commuters, then, refusing licence-plate surveillance on the way to work costs a few minutes at most. This is conditional on the assumed camera field of view: it is stable if the true radius is smaller than the 60 m assumed here and degrades if it is substantially larger, which Section 8.4 quantifies. It is the paper's central quantitative claim.

The distribution matters as much as the median. The mean overhead is 17.11 % against a median of 9.65 %, and the interquartile range runs from 0 % to 24.1 %: a quarter of commuters pay nothing at all, while a tail pays substantially. Figure 3 shows one such case.

6.3 Fifty states and the District of Columbia

State-level variation is extreme (Figure 4; every state's figures are tabulated in the repository). In Georgia 94.3 % of commuters pass a reader, a mean of 16.6 on the way to work; only 37.0 % can reach work past none, and avoiding costs a median of 13.6 min, an overhead of 44.2 %. In Illinois 91.4 % pass one, in California 89.9 %, in Florida 88.0 %. At the other end 19.8 % of Alaska's commuters pass one, 14.4 % of New Hampshire's, 11.8 % of Montana's and 7.9 % of Vermont's. In fourteen jurisdictions the median extra time is zero: for at least half of their commuters the avoiding route takes no longer than the usual one, in eleven of them because most commuters pass no reader at all. The registry moves some states far more than others, since it adds only where Flock operates and mappers had not reached: Arizona's share rises from 71.4 % to 85.8 %, Massachusetts's by eight points and Wisconsin's by seven, while New Hampshire, Vermont, Wyoming and Alaska do not move.

These are differences in the record at least as much as on the roads. Installation density runs from under five per 100,000 residents in Alaska and Vermont, and 5.1 in New Hampshire, to 160 in Georgia—an eight-fold spread between the 10th and 90th percentile states—and state exposure tracks it at $r = 0.87$. The departures from the line are what a statewide rate would produce: New York's commuters pass more cameras than its statewide density predicts, because commuting concentrates where cameras do while the density is averaged over the whole state; Kansas's pass fewer.

Whether evasion is cheaper per camera where cameras are denser, the data do not settle. On the map alone, across the 48 states and the District with at least 100 sampled commutes that evade

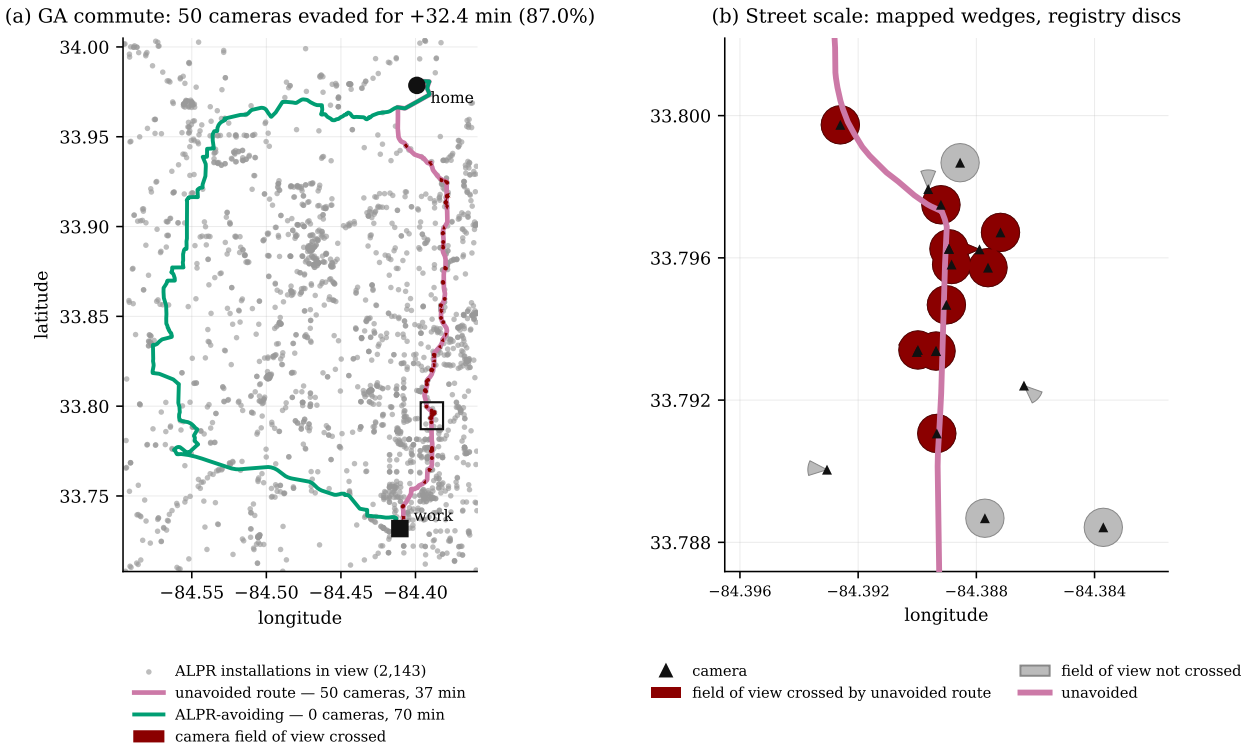


Figure 3: An illustrative high-exposure commute in metropolitan Atlanta—chosen for legibility, not typicality, and from the far tail of the cost distribution. (a) The unavoided route crosses 50 camera fields of view; the avoiding route crosses none, for 32.4 min (87.0 %). (b) At street scale, the mapped cameras are wedges aimed by their surveyed bearings and the registry's readers, whose direction is unknown, are discs; fields of view the unavoided route crosses are dark, those it misses grey, and they cost nothing. Map data © OpenStreetMap contributors.

a camera, the median price per camera falls weakly as cameras per kilometre rise, from 2.49 min in New Hampshire to 0.18 min in the District of Columbia ($r = -0.25$ against the logarithm of cameras per kilometre), but neither a Fisher interval, −0.49 to 0.04, nor a bootstrap over states, −0.58 to 0.33, excludes zero. On the merged map the correlation is weaker still, $r = -0.17$ (−0.43 to 0.12; −0.54 to 0.41). Half the registry's readers stand in four states the map already showed as heavily watched—Georgia, California, Texas and Florida—and there the price of evading each rises: Georgia's from 1.11 min to 1.32 min. The extremes hold, New Hampshire dearest and the District cheapest, but a tendency between them is not one we can claim.

6.4 Thirty-four months of mapping

Everything above describes the present, on the merged map. The registry records neither when a reader went up nor when one came down, so the past can be measured on the map alone. We re-asked the same questions of OpenStreetMap's camera map as it stood on the first of every month from January 2024, reconstructed exactly from its history (Section 4.4), and of the snapshots of 24 July, 22 September and 21 October 2026: the same 56,133 commutes on the same roads, each saved route re-scored against each month's cones, and, on the first of every quarter, every commute that passed a camera on that date routed again to avoid that date's cameras (Section 5.3). Table 3

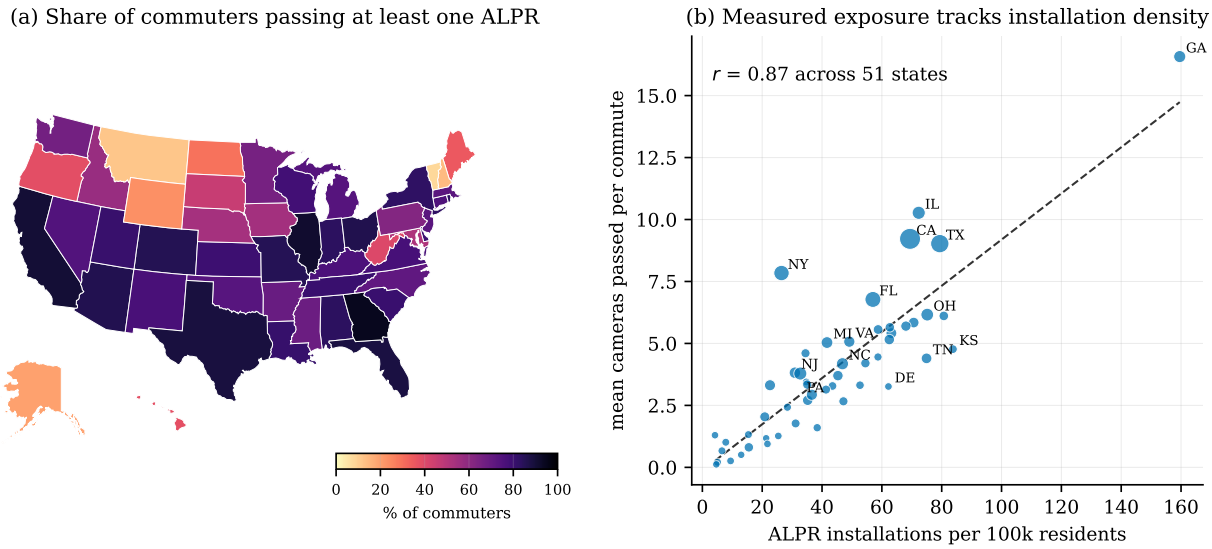


Figure 4: (a) Share of each state's commuters whose unavoided route passes at least one ALPR on the merged map. (b) Mean cameras passed against installations per 100,000 residents, points sized by commuters. Installation density alone tracks most of the variation between states ($r = 0.87$): from the record alone, a state with few cameras cannot be told from one whose cameras are unrecorded (Section 8.2).

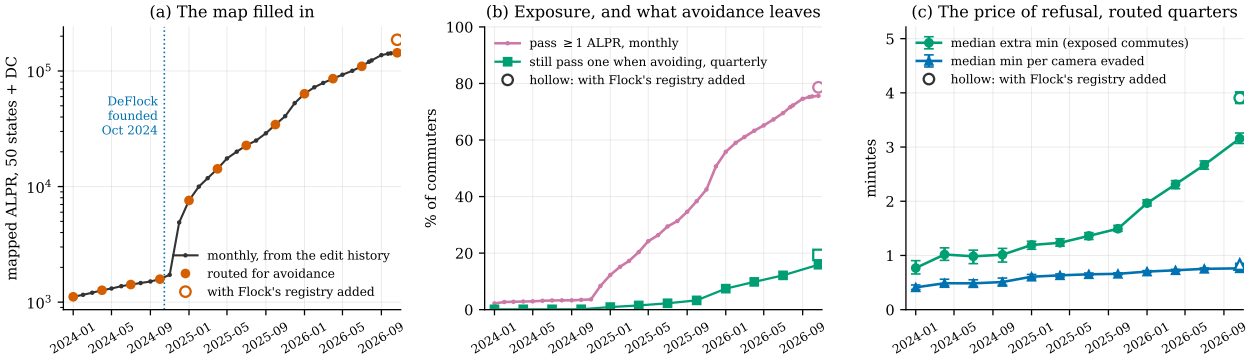


Figure 5: (a) Mapped ALPR in the fifty states and DC on the first of every month, from the edit history, with the routed dates marked. (b) Share of commuters passing at least one mapped reader on each monthly map (band: 95 % CI), and the share still passing one on the avoiding route, on the routed dates. (c) For commuters exposed on each routed date, the median extra minutes to avoid every camera, and the median minutes per camera evaded. Hollow markers: the October 2026 map with Flock Safety's registry added.

gives the routed dates, Figure 5 every month, and Figure 6 every state; the last row of the table and the hollow markers of the figure set the merged present beside the map's. Each date should be read as what had been *mapped* by then, not what had been installed.

The map filled in. On 1 January 2024 OpenStreetMap held 1,112 ALPR installations in the fifty states and the District of Columbia, four-fifths of them with no vendor tag, and twenty-six jurisdictions had none at all. It grew by about fifty a month until DeFlock was founded in October

Table 3: The same 56,133 commutes on the same roads against the camera map of each routed date, commuter-weighted. Exposure from re-scoring each saved route; avoidance from routing every commute exposed on that date, with the share reaching zero and the median extra time given for those commuters. The monthly maps between these dates are in Figure 5. The last row is the merged map: the October 2026 map with the registry's readers, routed in full, whose exposure columns come from the router rather than from re-scoring. Intervals come from the trend's own bootstrap and can differ from Table 2's in the last digit.

Map date	Mapped ALPR	Passing ≥1 (%)	[95% CI]	Mean passed	Exposed who reach zero (%)	Extra min, exposed	Min per camera
2024-01-01	1,112	2.2	[2.1–2.4]	0.05	95.2	0.77	0.41
2024-04-01	1,267	2.9	[2.7–3.1]	0.08	95.8	1.02	0.49
2024-07-01	1,420	3.3	[3.0–3.5]	0.08	96.2	0.98	0.49
2024-10-01	1,581	3.5	[3.2–3.7]	0.09	96.1	1.01	0.51
2025-01-01	7,583	12.3	[11.9–12.7]	0.35	92.5	1.19	0.61
2025-04-01	14,238	20.4	[19.9–20.9]	0.59	92.7	1.23	0.63
2025-07-01	22,690	29.4	[28.9–30.0]	0.94	92.3	1.36	0.65
2025-10-01	34,383	38.4	[37.8–39.0]	1.33	91.4	1.49	0.66
2026-01-01	63,522	55.8	[55.2–56.4]	2.44	86.7	1.96	0.70
2026-04-01	85,845	63.3	[62.7–63.8]	3.14	84.5	2.31	0.73
2026-07-01	109,848	69.5	[69.0–70.1]	4.08	82.6	2.67	0.75
2026-10-21	143,721	75.6	[75.1–76.1]	5.17	79.0	3.16	0.76
+ registry	186,247	78.6	[78.2–79.1]	6.21	75.4	3.91	0.83

2024, and then abruptly: 1,724 on 1 November, 4,894 on 1 December, 7,583 on 1 January 2025. From there the map grew by a median 13.8 % a month—22,690 by July 2025, 63,522 by January 2026, 109,848 by July, and 143,721 on 21 October 2026, a 129-fold increase in thirty-four months (Figure 5a). As a rate the pace eased, to a median 9.2 % a month over the first eight months of 2026, while in cameras it quickened, from about 4,700 a month during 2025 to 9,200 in 2026; then it slowed sharply, to 5,121 (3.7 %) in September and 1,416 in the three weeks to 21 October. By July 2025 only Hawaii, Vermont and Alaska had no mapped reader; by January 2026 only Alaska; on 1 July 2026 none, though Alaska had none again on 24 July and 1 September, and 31 by 21 October. Flock's share of the mapped network went from 17 % to 71 % during 2024 and has stayed near four-fifths since, while the untagged share fell from 81 % to 5 % and the named non-Flock vendors rose from 1 % to 15 % as contributors tagged them.

Exposure followed the map, almost in proportion. The share of commuters passing at least one mapped reader crossed a tenth in January 2025, a quarter in June, a half in December, and stands at three-quarters (75.6 %) today; the mean went from 0.05 cameras to 5.17, and adding the registry takes them to 78.6 % and 6.21. Against the number of mapped cameras the elasticity is 0.90 across the thirty-seven maps. Per camera mapped, exposure has fallen by about a quarter since January 2024, from 0.049 to 0.036 encounters per commute per thousand cameras: the cameras mapped later lie on commute routes slightly less often than those mapped first, but not much less. Exposure kept pace while the map grew: the two months between the July and September 2026 snapshots brought 17.3 % more mapped cameras and 16.7 % more encounters, and the monthly curve of Figure 5b rose by about two points a month from March to September 2026. It slowed with the map, by 0.8 points in September and 0.2 in the three weeks to 21 October; one slow month

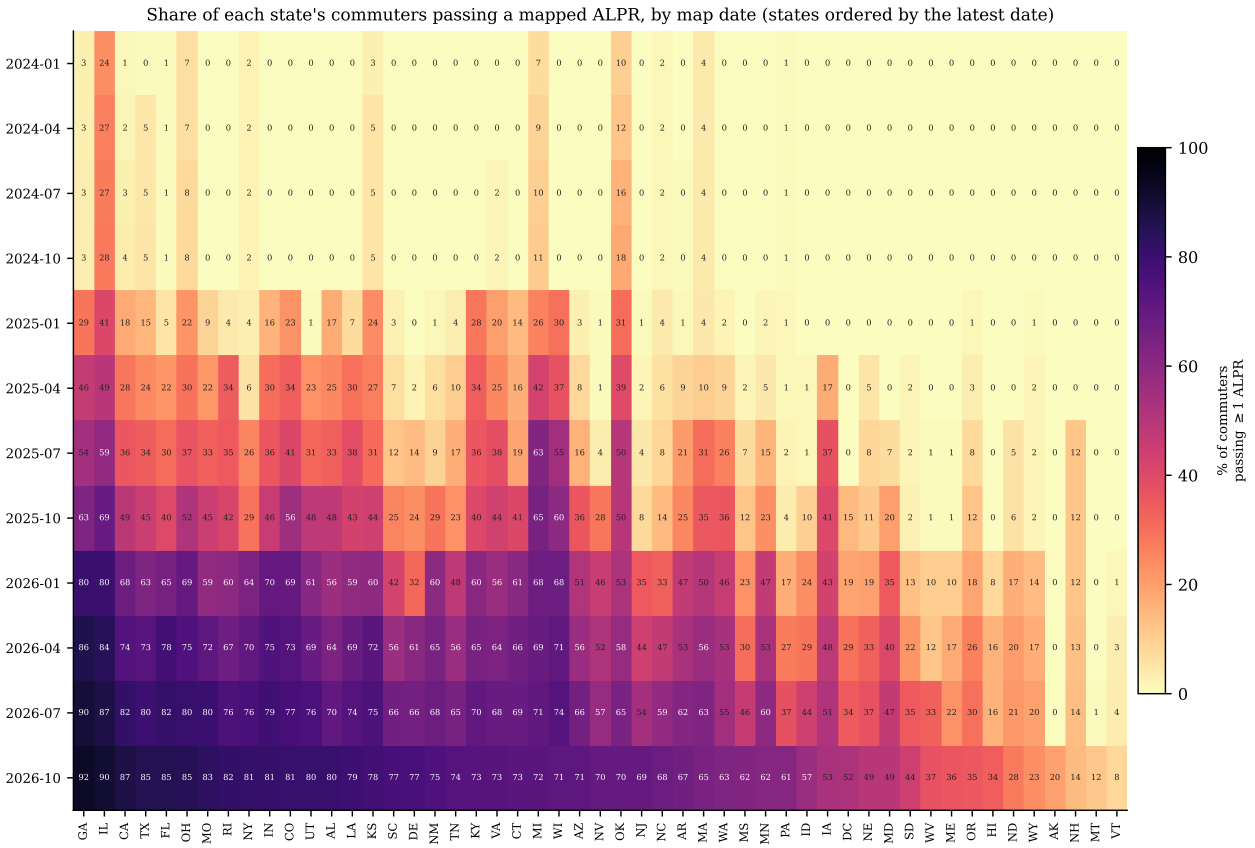


Figure 6: Share of each state's commuters passing at least one mapped ALPR on each routed map, states ordered by the latest.

cannot say whether mapping is nearing what is installed or has merely paused.

The states filled in at different times. In January 2024 only Illinois (23.6 %) and, barely, Oklahoma (10.1 %) had more than a tenth of commuters passing a mapped reader; by January 2025 fifteen jurisdictions did, by July 2025 thirty-one, by January 2026 forty-seven, and today fifty, all but Vermont (Figure 6). In July 2025 Michigan's commuters were the most exposed of any state's (62.7 %), while Pennsylvania (2.2 %), Nevada (3.8 %), New Jersey (4.5 %) and North Carolina (8.2 %) would have looked nearly camera-free; all four are between 61 % and 70 % today. A state's rank on the January 2025 map predicts its rank today imperfectly (Spearman $\rho = 0.79$ on mean exposure; from January 2024, when 37 jurisdictions had none, 0.53; from January 2026, 0.88). Nor does the map only grow. Wisconsin and Iowa each gained cameras on net between the July and October 2026 maps yet lost 3.5 and 3.1 points of exposure: 223 and 127 of their July cameras left the map, deleted or retagged, and the cameras newly mapped added less exposure than was lost.

Refusal got dearer, and not only because there was more to refuse. On the sparse maps of 2024 a commuter who passed a mapped camera could almost always get around it: on the first, 95.2 % [93.0–97.1] of exposed commuters could reach zero, for a median of 0.77 min, and through 2024 the share held at 95 % to 96 % for about a minute. On the October 2026 map 79.0 % [78.4–79.6] can, for a median of 3.16 min, and with the vendor's registry added 75.4 % [74.7–76.0], for 3.91 min

[3.81–4.01] (Table 3; Figure 5c). The registry moves the present about as far as the preceding months of mapping did: the share reaching zero by as much as the sixteen weeks from the July map, the median detour by most of the rise since April's map (0.75 min of 0.85 min). The quarterly maps show the shape of the change. The share reaching zero held at 95 % to 96 % through 2024, stepped down to about 92 % with the surge of mapping that followed DeFlock's founding and held there through 2025, and has fallen on every routed map since October 2025: 91.4 %, 86.7 %, 84.5 %, 82.6 % and 79.0 %, each interval clear of the last. The cost did not step; it climbed. From July 2024 on, the median extra time for an exposed commuter rose on every routed map, from 0.98 min to 3.16 min. Most of that rise is the number of cameras on each trip, but the price of evading each one rose too, from 0.41 min [0.37–0.46] to 0.76 min [0.75–0.78], and it too rose on every routed map from July 2024. Two things moved it, and a fixed sample can separate them. One is who was exposed: each fuller map reached commuters the sparser ones had not, on other roads with other alternatives. The other happened on the same trips. Followed from the July 2025 map to October 2026's, the 12,642 commutes exposed in July came to pass 9.9 cameras on average instead of 3.2; the share of them that can reach zero fell from 92.3 % to 72.6 %, and the price per camera rose from 0.65 min [0.63–0.68] to 0.81 min [0.78–0.83]. Every cohort shows it, and for nearly two years the rise grows with the time a cohort is followed: the commutes exposed on the July 2026 map pay 2 % more per camera sixteen weeks later, too little to distinguish; April 2026's pay 7 % more, January's 11 % and October 2025's 20 %, each interval clear of its starting one. The seven cohorts followed for fifteen months or more pay 24 % to 38 % more, each clear of its starting interval. As more of a trip's cameras, and of its alternatives', are mapped, each costs more to avoid. Over the two months between the July and September 2026 snapshots the change is too small to see, which is why the fifteen-state comparison below finds the price per camera unchanged.

Fifteen states, fully re-routed. A step of the trend can be checked against a comparison that re-routed everything. In September, for the fifteen states of the study's earlier rounds, the same 17,580 commutes were routed in full against the July and September 2026 snapshots on the same road extract, and every baseline route was identical to the metre and the second. The September snapshot still held the erroneous Chicago nodes of Section 4.4; both sides of the comparison used it. Camera encounters grew by 18.3 % against 18.7 % more mapped cameras in those states, the median price of evading one stayed at 0.78 min to the second decimal, and Pennsylvania's mean exposure rose 88.5 % in two months.

What follows. Two things. Absolute exposure figures from a volunteer map are a lower bound that has not fallen from one month's map to the next and will keep rising; one vendor's records alone raise them again, exposure by a fifth and the price of refusal by two-fifths. Any study built on crowdsourced surveillance data dates, and should say when its map was drawn. And the price of refusal dates as well: it has risen on every routed map since July 2024, per camera as well as in total, and today's figures describe a record that is still filling in.

7 Discussion

7.1 Refusal is cheap, and that is the policy-relevant fact

The strongest result here is also the simplest: 80.6 % of American commuters can reach work past no known licence-plate reader, and the avoiding route is a median of 2.29 min longer—3.91 min for commuters with a camera to avoid. A common intuition about these networks is that they are, or

soon will be, comprehensive enough that avoidance is futile, so that the privacy loss is unavoidable rather than chosen. At current density that is false for four commuters in five. It has been false on every map since 2024, but it is becoming less so: as the map filled in, the share of exposed commuters who can reach zero fell from 95 % to 79 %, and to 75 % once the vendor's registry is added, while the median cost for those with a camera to avoid quadrupled on the map alone and rises by a further quarter with the registry (Section 6.4).

Cheap refusal also presumes a router that knows where the cameras are. The app built on this study's cone geometry avoids the volunteer map alone (Section 3). Routes planned around the map alone are clean of it for 84.1 % of commuters, but for 34.1 % [33.4–34.7] of them the route still passes a reader that only the vendor's registry records, and for 42.8 % of those whose usual route passes a mapped camera; 55.5 % of commuters get a route clean of both. Avoidance is only as complete as the record it avoids.

The corollary cuts the other way. If avoidance is this cheap, it is also cheap for people with reason to evade legitimate investigation, which bounds how much investigative value a fixed-camera network can deliver against a motivated subject while it continues to read every unmotivated one.

7.2 Where refusal stops being cheap

The national figures average over states whose commuters have almost nothing to avoid and one where avoidance has largely stopped working. In Georgia, the most densely recorded state, 94.3 % of commuters pass a reader, the average commute passes 16.6, and only 37.0 % can reach work past none; avoiding costs a median of 13.6 min, an overhead of 44 %. On the map alone Georgia's commuters could reach zero in 53.2 % of cases, for 7.8 min: the 6,919 readers the vendor's registry adds there turn refusal into something most of them cannot complete. Each camera there also costs more to avoid than nationally, 1.32 min against 0.83 min, which is the cohort finding of Section 6.4 in one place: the more of a trip's cameras, and of its alternatives', are on the record, the dearer each is to evade.

On the map alone the opposite pattern appears: evasion looks cheapest per camera where cameras are densest, a pattern that would rest on road-network structure, which OpenStreetMap maps close to completely, and so might be expected to survive the map's gaps in coverage. The vendor's registry is a test of exactly that, and the pattern does not survive it: across states it weakens to a correlation neither interval distinguishes from zero (Section 6.3). Which roads are watched is itself a question of coverage, so even a result built on the road network moves when the record of cameras does.

8 Limitations

8.1 Tract centroids are not addresses

Every route runs centroid to centroid; real commutes begin in a driveway. In large or irregular tracts the centroid can sit far from any residence. This adds noise in both directions and is the main reason we report distributions rather than point estimates. It also costs coverage: where a tract's internal point lies too far from any road to snap to, the commute cannot be routed at all, which removes a fifth or more of the sample in Alaska, Hawaii, Montana and Wyoming (Section 5.5).

8.2 Crowdsourced camera data is non-randomly missing

Absolute exposure figures remain a **lower bound**, now a tighter one. Like volunteered geographic data generally [36, 38], the map is least complete where contributors are absent, and there cameras are missing from the map but present on the pole. Against Flock Safety's own transparency portals the map alone holds, at the median, as many cameras as the 1,096 municipal police departments report operating (a ratio of 1.00; in total 29,489 mapped inside their limits against 31,865 reported), small towns being the least completely mapped [20]; against the vendor's registry it holds 64.6 % of in-service plate readers (Section 4.2). The merged map closes that gap for one vendor, as of December 2025. What it cannot close: other makers' unmapped cameras, which the registry does not cover; Flock devices installed since December 2025, which the registry predates and the map holds only where mappers have reached them; and whatever the registry itself omits—Flock has said it operates more than 120,000 cameras nationwide [51], against the 126,389 in-service plate readers the registry carries, so that gap at least appears small. The trend of Section 6.4 is the best evidence of how far the map itself still has to go.

8.3 One vendor's registry, as it stood in December 2025

The registry closes the map's largest gap for one vendor at one moment, and inherits both limits. Its positions are the vendor's records rather than a survey: where a mapped node stands on the same device the two agree to a median of 5.6 m, but nothing checks the 42,811 readers no mapper has reached. Its statuses are December 2025's, ten months before the map's snapshot: the removals since that an agency, the press or a mapper recorded are taken out (Section 4.2), but one nobody recorded is still counted, and the 34,325 the registry then listed as planned, many presumably installed since, are not. Its readers carry no bearing, so each is a disc rather than a wedge, which the directed scoring of Section 5.2 brackets. The merged figures can therefore err either way: the discs and any unrecorded removals overstate, while the planned readers installed since and other makers' unmapped cameras (Section 8.2) understate.

8.4 Cone geometry: swept, and one result depends on it

The 60 m radius was chosen by reasoning—a pole set back from the kerb needs a cone long enough to reach the carriageway it watches—and the 45° default span from the 82 % of explicit arcs that span exactly that. Neither was measured. We therefore swept the radius over $\{30, 45, 60, 90\}$ m, rebuilding the map's cone geometry at each and re-scoring all 56,133 routes of the map-alone run (Table 4).

Table 4: Cone-radius sensitivity, commuter-weighted. Exposure is mildly sensitive to the assumption; the avoidance result is not, until the radius grows by half. At the assumed 60 m the clean share is 83.7 % rather than the map-alone 84.1 % of Table 2 because the sweep re-scores stored route geometry (Section 5.3).

Radius	Mean cameras	vs 60 m	Avoided routes still clean	Residual
30 m	3.84	0.74×	87.0 %	5.5 %
45 m	4.64	0.90×	85.5 %	5.3 %
60 m	5.17	assumed	83.7 %	5.4 %
90 m	5.92	1.14×	**54.7** %	21.8 %

Two things follow, and they point in opposite directions. Exposure is robust: halving the radius removes a quarter of it and 90 m adds 14 %, so Section 6.1 does not rest on the choice. The avoidance headline is conditional on it, and from above only: routes planned against 60 m cones stay clean when re-scored at 45 m and 30 m, but at 90 m the share reaching zero falls from 83.7 % to 54.7 % (−29.0 points, 95 % CI [−29.6, −28.4]) and residual exposure quadruples. That is a lower bound on what avoidance could achieve at 90 m, not an estimate of it: these routes were planned against 60 m cones and only re-scored, a router given wider cones would find different clean routes, and a full re-route at each radius—most of a day apiece—has not been done. But if real fields of view are half again as long as assumed, more than a third of the routes counted as clean here are not. The sweep re-scores the map's wedges; the registry's discs share the radius and have no span to sweep, and the directed scoring of Section 5.2 is their sensitivity check.

8.5 A single avoidance strength

All results use the strongest setting, 0.01: a watched road costs a hundred times its usual weight, close to but short of prohibition, so "zero cameras" means the optimiser found a clean path, not that one was guaranteed to exist. Intermediate settings are the practically interesting ones for a shipped application, and in informal tests whether one took effect depended on local camera density; a single slider-to-multiplier mapping is unlikely to serve.

8.6 Sample scope

All fifty states and the District of Columbia, but within-state commutes only (`od_main`), so cross-border metropolitan areas—New York/New Jersey, Philadelphia, Washington, Kansas City, St Louis, Portland—are systematically truncated at the state line. Several are exactly the dense multi-jurisdiction areas where the question is most pointed. The District is the extreme case: most people who work there live in Maryland or Virginia, so its sample describes only commutes with both ends inside the District. Alaska's flows are from 2016 and Michigan's from 2021, the latest LODES publishes; the other 49 jurisdictions' are from 2022.

LODES pairs a home with a job's establishment, and one pair in eight in the sample lies more than 100 km apart (Section 4.6). Those pairs make the tail, and they raise the price of refusal more than the share exposed: within 100 km, 76.1 % [75.6–76.6] of commuters pass a reader rather than 78.6 %, but avoiding costs a median 1.70 min [1.65–1.76] rather than 2.29 min, and the mean commute passes 4.90 cameras rather than 6.21. The registry's effect is much the same at that range: on the map alone the figures are 72.8 % and 1.23 min.

8.7 Past maps record mapping, not deployment

The trend of Section 6.4 is a trend in what had been mapped. The edit history gives the day each camera entered the map to the second; it does not give the day the camera went up, which OpenStreetMap records for 49 of the 143,721. A camera first mapped in 2026 may have stood since 2021, and the 129-fold growth of the map cannot be read as a rate of deployment. Absolute figures date accordingly. The relative claims are the ones to carry forward, and they held on every map: the cheapness of refusal—under a minute per camera evaded on all twelve routed maps, though rising on each since July 2024—and the near-proportionality of measured exposure to the map across all thirty-seven. What the maps cannot establish is that the current one is complete, and the merged map of the present has no past: the registry's record dates are not installation dates (Section 4.2).

8.8 Free-flow travel times

Routing uses free-flow speeds with no traffic, time-of-day, or turn-restriction penalties, so both travel times are optimistic. The *difference* between them—the quantity actually reported—is more robust than either absolute, since both share the bias.

9 Ethics and dual use

The study uses only public data. The camera locations are volunteers' observations from public places, already published in OpenStreetMap under an open licence, and the vendor's device records, already published by an independent researcher [55], which we use in aggregate and do not redistribute; the commute flows and neighbourhood attributes are Census Bureau aggregates, published with the Bureau's confidentiality protections and attached to census tracts rather than to people; the portal counts are published by the police departments themselves. No individual is identified, no one's travel was observed, and the paper adds no camera location to the public record.

The routing engine is dual-use, and we weighed that before publishing it. It lets anyone decline to be photographed on the way to work, including someone with reason to evade a lawful investigation. Three considerations decided the matter. The capability is not new: the camera map is public, and several services already route drivers around it [3, 11, 25]. The result it rests on bounds the investigative value of fixed cameras against a motivated subject whether or not the tool exists (Section 7); what such a network reliably records is the unmotivated majority. And the measurement itself—how much surveillance an ordinary trip passes, what declining it costs, and how fast both are growing—is information a public debate about the technology needs and has not had. The app is released under GPLv3 and the server and study code under the MIT licence, and the study's routing server is not exposed to the internet (Section 3).

10 Reproducibility

Code, configuration, and the derived outputs behind every table and figure—per-commute results on every routed date, per-state tables, and every analysis report quoted in the text—are in the project repository, `https://github.com/KaraZajac/TRAMES`, under `research/`. Fixed seed `20260725` throughout, for both sampling and bootstrap. All external data sources are public and require no API key. Beyond the routing server, the analysis needs Python with NumPy, Shapely and Matplotlib; pyosmium for the road sample and the history dump; and pyogrio, which bundles GDAL, only to read the HPMS geodatabase. The repository's README names the script behind each step, from the camera fetch and the edit-history extraction through sampling, routing, the past-date runs and the analyses, to these figures and tables.

The 19.5 GB OSM extract, the four 14.8 GB graph caches, the 164 GB OSM full-history dump, the 1.3 GB of TIGER/Line boundaries and the 5.6 GB HPMS geodatabase are not redistributed; the acquisition and import scripts rebuild all of them. The transparency-portal counts behind the coverage check come from a live service and are kept as retrieved, in `research/data/validation/`. The registry snapshot is not redistributed: it is available from its publisher [55], and the scripts that read it, merge it with the map and build its discs are in the repository. Commuter weights come from `research/out/sampling_frame.json` and `sample_draws.csv`, written by the sampler.

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A Registry readers removed by the cutoff

Flock Safety's registry describes its devices as they stood in December 2025. A reader it lists as in service is marked removed (Section 4.2) where a removal by 21 October 2026 is on record, in one of two ways. Table 5 lists the 32 places where the operating agency, or reporting on it, says its readers were taken down, 417 records in all. Each place's in-service records were matched to the agency by the registry's own labels and counts; where a source gives no count, it says that all of the agency's readers came down. Two of the sources describe theft rather than removal (Winona, and four of Hubbard's six), and none reports the cameras replaced; Arlington County's reporting counts 26 plate readers where we drop 27 records, one of which may be Alexandria's. A further 572 records lost, between September and the cutoff, the mapped node they were paired with to a deletion whose changeset gives the camera's removal as the reason (*device removed*, *cameras removed by vendor*, or a citation of the Florida transportation department's order). Every record, with its source or changeset, is in `research/data/registry_removals_2026-10.json`.

Table 5: Registry readers marked removed on the evidence of the operating agency or the press, by place: records dropped, when the source says the readers came down, and the source.

Place	Records	Removed	Source
Seminole County (sheriff), FL	101	by 30 Sep 2026	Howard [39]
St. Johns County, FL	65	Sep 2026	Bazley [4]
El Cerrito, CA	40	about 21 Aug 2026	Lee [45]
Flagstaff, AZ	32	by 19 Feb 2026	City of Flagstaff [7]
Harrisonburg, VA	28	Aug 2026	The Northern Virginia Daily [70]
Arlington County, VA	27	Oct 2026	Egitto [17]
Itasca County, MN	15	by 10 Sep 2026	Krab [43]
Syracuse, NY	13	by 15 Jul 2026	McCarthy [52]
Bloomington, IN	11	by 7 Oct 2026	Knapp et al. [42]
Moffat County, CO	10	Sep 2026	Meadows [54]
Santa Cruz, CA	8	by 9 Mar 2026	Santa Cruz Local [64]
South Portland, ME	7	by 22 Jun 2026	City of South Portland [8]
Hubbard, OH	6	Aug 2026	Ross [63]
Winona, MN	6	29 Jul–1 Aug 2026	Longworth [47]
Christiansburg, VA	5	by 23 Sep 2026	Forrest [26]
Columbia Heights, MN	5	Jun 2026	Bagnoli [2]
Staunton, VA	5	11 Feb 2026	Graham [34]
Erlanger, KY	4	by 27 Aug 2026	Andres [1]
Cudahy (a private gun range), WI	3	4 Aug 2026	Dysart [14]
Hillsborough, NC	3	by 3 Dec 2025	Town of Hillsborough [71]
Stow, MA	3	by 26 Aug 2026	Ellement [19]
Tampa, FL	3	by 30 Sep 2026	City of Tampa [9], Williams [74]
Verona, WI	3	by 20 Apr 2026	Lehr [46]
Augusta County (Fishersville), VA	2	25 Sep 2026	Dobrin [12]
Evanston, IL	2	Mar 2026	Harrison [37]
Mayflower, AR	2	by 26 Aug 2026	Williams [75]
Nantucket, MA	2	17 Sep 2026	McGrady [53]
Racine County (sheriff), WI	2	4 Sep 2026	FOX6 News Digital Team and Polcyn [28]
Crystal, MN	1	by 9 Sep 2026	FOX 9 [27]
Liberty Hill, TX	1	15 Sep 2026	Gonzalez [32]
Ocean Shores (county sheriff), WA	1	by 30 Jul 2026	The Daily World [69]
Washington Island, WI	1	Jan 2026	Small [67]
All 32 places	417		